

DFP40203 PYTHON PROGRAMMING

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LABORATORY EXERCISE

TOPIC 4.2 UNDERSTAND EXCEPTION

1. Write a program that asks the user to enter two numbers and divides the first number by the second number. Handle the ZeroDivisionError exception and print an appropriate message to the user.

Coding

```
# Question 1
def divide(FNumber, SNumber):
    try:
        print(f'{FNumber}/{SNumber} is {FNumber / SNumber}')
    except ZeroDivisionError :
        print("You put the zero(0) Number")

divide(10,0)
```

Output

```
You put the zero(0) Number
```

2. Write a program that prompts the user to enter a filename. If the file exists, the program should display its contents. If the file does not exist, the program should handle the FileNotFoundError exception and print an appropriate message to the user.

Coding

```
try:
    file = open("my_letter.txt", "r")
    print(file.read(10))
except FileNotFoundError:
    print("Cannot found the File.")
```

Output

```
Hai my nam
```

Coding

```
try:
    file = open("myLetter.txt", "r")
    print(file.read(10))
except FileNotFoundError:
    print("Cannot found the File.")
```

Output

```
Cannot found the File.
```

3. Write a function that takes a list of numbers as an argument and returns the sum of the even numbers in the list. If the list contains non-numeric values, the function should handle the ValueError exception and print an appropriate message to the user.

Coding

```
def sum():  
    try:  
        a = int(input("Enter a number: "))  
        b = int(input("Enter a number: "))  
        add = a + b  
        print(add)  
    except ValueError:  
        # Handle the exception  
        print("Invalid input. Please enter a valid integer.")  
  
sum()
```

Output

```
Enter a number: 12  
Enter a number: 23  
35
```

Output

```
Enter a number: 12  
Enter a number: r  
Invalid input. Please enter a valid integer.
```

4. Write a program that reads a text file and counts the number of occurrences of a particular word. Handle the FileNotFoundError and IOError exceptions and print an appropriate message to the user.

Coding

```
try:  
    filename = input("Enter the filename: ")  
    word = input("Enter the word to search for: ")  
  
    file = open(filename, 'r')  
    text = file.read()  
  
    count = text.count(word)  
    print(f"The word '{word}' appears {count} times in the file '{filename}'.")  
except FileNotFoundError:  
    print(f"Error: file '{filename}' not found.")  
except IOError:  
    print(f"Error: unable to read file '{filename}'.")
```

Output

```
Enter the filename: filesaya.txt
Enter the word to search for: malaysia
The word 'malaysia' appears 1 times in the file
'filesaya.txt'.
```

Output

```
Enter the filename: fileme.txt
Enter the word to search for: www
Error: file 'fileme.txt' not found.
```

5. Write a program that asks the user to enter a number between 1 and 10. If the number is outside of this range, the program should handle the ValueError exception and print an appropriate message to the user.

Coding

```
class InvalidNumber(Exception):
    pass
try:
    number=int(input('Enter the number between 1 to 10:'))
    if number > 10 :
        raise InvalidNumber
    else:
        print("Done you put ",number)
except ValueError :
    print("Invalid Value : The value must in integer not string")
except InvalidNumber :
    print("Invalid Number : The terms of number is between 1 to 10")
```

Output

```
Enter the number between 1 to 10:5
Done you put 5
```

Output

```
Enter the number between 1 to 10:a
Invalid Value : The value must in integer not string
```